Report R2 — Unitary sweep and fragmentation scaling

Krylov sector analysis of quantum cellular automata (Tier 1a/1c)

QCA Sector Analysis project (Tier 1)

2026-07-12 (engine_version `1a.1`)

Abstract

We sweep all sixteen unitary quantum elementary cellular automata over chain lengths $N = 6 \ldots 18$, both boundary conventions, and classify each by the scaling of its Krylov-sector structure. Sector sizes are exact; the growth of the number of sectors and of the largest sector $D_{\max}$ is fit by BIC model selection ($\ln y = c\,[+\alpha \ln N]\,[+\kappa N]$). Seven rules are effectively ergodic under `obc0` and nine under `pbc`; that verdict originally rested on $N = 6$ alone, and §4 re-establishes it over $N = 6 \ldots 13$ with the full sector decomposition, where it turns out to be exactly N-independent and derivable from the rule encoding. The remaining rules fall into three fragmentation families: exponential (156/198, 108, 201), a domain-wall family with a linearly growing sector count but an extensive $D_{\max} = 2^{N-1}$ (60/102, `obc0` only), and 150/105 with central-binomial sectors. We highlight two boundary-sensitive phenomena and one broken symmetry.

1 Method

Each unit (rule, N, bc) is computed by the exact Tier-1a engine (Report R1) and cached as one JSON line in `results/`. Unitary sectors are the weakly connected components of the one-cycle transition graph (extracted by union–find; see R1). A rule is flagged *effectively ergodic at N* when one sector exceeds 90% of the 2^N states; the sweep then stops increasing N for that rule/bc. Growth classes are assigned to the series $y(N) \in \{\#\text{sectors}, D_{\max}\}$ by BIC selection among constant / power-law / exponential models, with the $\alpha \ln N$ prefactor always carried when reporting the exponential rate κ (so that $\sqrt{N}$-type binomial prefactors do not bias κ).

2 Results

Table 1 lists every computed rule/bc with its sector count and $D_{\max}$ at the largest N reached and the fitted growth classes. Figures 1–3 show the scaling.

3 A caution on reading bases off these fits: rule 156

The growth-base map invites reading a constant off each point, and 156/198 is the case where that fails. Its $D_{\max}$ series (`obc0`) is

$$6,\ 9,\ 12,\ 16,\ 20,\ 27,\ 36,\ 48,\ 64,\ 81,\ 108,\ 144,\ 192 \qquad (N = 6 \ldots 18),$$

and it satisfies *both* $a(N+4) = 3a(N)$ and $a(N+5) = 4a(N)$ with exactly the same two exceptions ($N = 6, 10$). The implied bases are $3^{1/4} = 1.31607$ and $4^{1/5} = 1.31951$ — they differ by 0.26%, and thirteen terms cannot choose between them. No fit should be allowed to.

W	tuple	bc	$N_{\max}$	#sec	$D_{\max}$	#sec growth	$D_{\max}$ growth (base)
150	`IVVI`	obc0	20	11	352716	polynomial	exponential (2.022)
60	`IIVV`	obc0	21	22	1048576	polynomial	exponential (2.000)
102	`IVIV`	obc0	21	22	1048576	polynomial	exponential (2.000)
105	`VIIV`	obc0	21	11	705432	polynomial	exponential (1.984)
201	`VIII`	obc0	21	97229	28657	exponential	exponential (1.618)
108	`IIIV`	obc0	21	170625	10946	exponential	exponential (1.617)
156	`IIVI`	obc0	21	28657	432	exponential	exponential (1.292)
198	`IVII`	obc0	21	28657	432	exponential	exponential (1.292)
204	`IIII`	obc0	21	2097152	1	exponential	constant
51	`VVVV`	obc0$^{\dagger 6}$	–	–	–	ergodic	ergodic
54	`IVVV`	obc0$^{\dagger 6}$	–	–	–	ergodic	ergodic
57	`VIVV`	obc0$^{\dagger 6}$	–	–	–	ergodic	ergodic
99	`VVIV`	obc0$^{\dagger 6}$	–	–	–	ergodic	ergodic
147	`VVVI`	obc0$^{\dagger 6}$	–	–	–	ergodic	ergodic
153	`VIVI`	obc0$^{\dagger 6}$	–	–	–	ergodic	ergodic
195	`VVII`	obc0$^{\dagger 6}$	–	–	–	ergodic	ergodic
105	`VIIV`	pbc	20	13	369512	constant	exponential (1.994)
150	`IVVI`	pbc	20	13	369512	polynomial	exponential (1.940)
108	`IIIV`	pbc	21	134643	24476	exponential	exponential (1.619)
201	`VIII`	pbc	21	134643	24476	exponential	exponential (1.619)
156	`IIVI`	pbc	21	24477	324	exponential	exponential (1.301)
198	`IVII`	pbc	21	24477	324	exponential	exponential (1.301)
204	`IIII`	pbc	21	2097152	1	exponential	constant
51	`VVVV`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
54	`IVVV`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
57	`VIVV`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
60	`IIVV`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
99	`VVIV`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
102	`IVIV`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
147	`VVVI`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
153	`VIVI`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic
195	`VVII`	pbc$^{\dagger 6}$	–	–	–	ergodic	ergodic

Table 1: Per-rule fragmentation scaling (unitary rules). $^{\dagger}N$ marks the N at which a rule first becomes effectively ergodic for that boundary convention. Rules reduced by reflection are represented once where noted.

The combinatorial derivation does choose. Writing the chain as "rooms" of length l separated by domain walls, a room contributes $l+1$ states and consumes $l+2$ sites, so $D_{\max} = b^N$ with

$$b(l) = (l+1)^{1/(l+2)}, \tag{1}$$

whose saddle point $\mathrm{d}\ln b/\mathrm{d}l = 0$ gives the transcendental condition $\ln x = 1+1/x$ with $x = l+1$, i.e. $l \approx 2.5911$. The two integer neighbours of that optimum are precisely the two candidates above, and $l = 3$ is the larger:

$$b(2) = 3^{1/4} = 1.31607 \ < \ b(3) = 4^{1/5} = 1.31951.$$

So the asymptotic base is $4^{1/5}$, and at finite N the largest sector is a *mixture* of rooms of length 2 and 3 — which is exactly why neither pure recurrence is exact over the computed range. This is the one base in the report that comes from theory rather than from the data, and it should stay that way.

A second, purely numerical trap sits on the same rule. The BIC model $\ln y = c+\alpha\ln N+\kappa N$ is the right tool for the growth *class*, but its $\alpha \ln N$ term absorbs part of the growth and makes

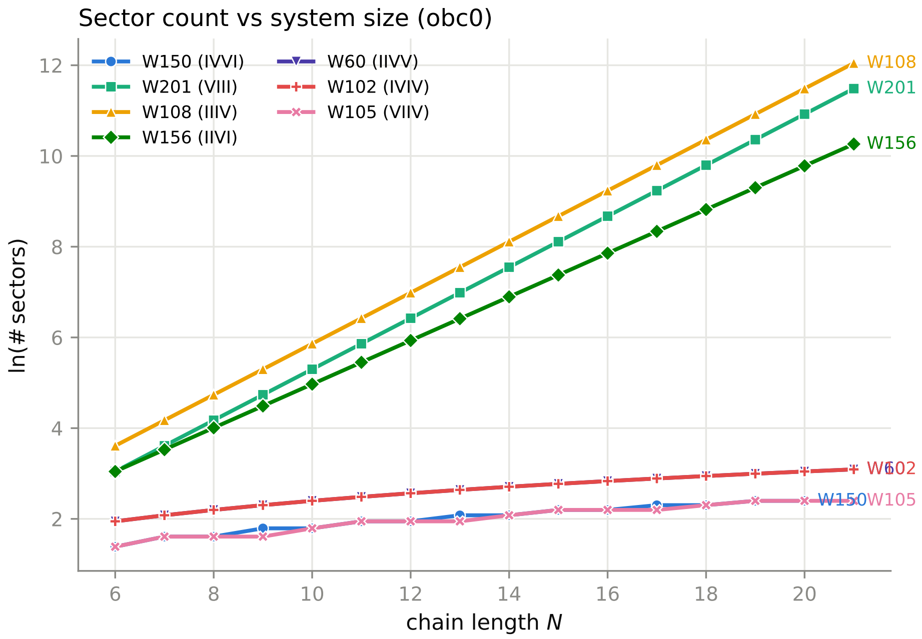

Figure 1: Number of Krylov sectors vs N (`obc0`), log scale. Exponential fragmentation (156, 108, 201) vs. linear/central-binomial sector counts (60/102, 150/105).

κ a biased rate estimator: it reports base 1.281 for 156/198, far enough below $4^{1/5}$ to look like a different constant. Rates in the growth-base map are therefore taken from the two-parameter fit $\ln y = c + \kappa N$, or from an exact recurrence where one exists.

4 The ergodic rules are ergodic at every N, not just at $N = 6$

The ergodicity verdict as originally produced rested on a single system size. The sweep aborts a unit the moment one component exceeds 90% of 2^N and then stops increasing N for that rule, so an ergodic-flagged unit had exactly one record, at $N = 6$ — and it carried no sector data at all, only the flag. Six sites is far too small to distinguish genuine ergodicity from a finite-size accident, so this claim was re-established from scratch: every ergodic-flagged unit was recomputed with the abort disabled (`run_rule --no-ergodic-abort`), which performs the *full* decomposition and reports how many sectors there really are, over $N = 6 \ldots 13$.

The outcome is completely rigid. Across all sixteen units and all eight sizes, only two structures occur — one sector of size 2^N, or one sector of size $2^N - 1$ together with a single frozen basis state — and *each unit gives the same structure at every* N. Nothing drifts, nothing crosses over, and no small sector appears at any size.

rule	tuple	`pbc`	`obc0`
51	`VVVV`	$\{2^N\}$	$\{2^N\}$
57	`VIVV`	$\{2^N\}$	$\{2^N\}$
99	`VVIV`	$\{2^N\}$	$\{2^N\}$
54	`IVVV`	$\{2^N - 1, 1\}$	$\{2^N - 1, 1\}$
60	`IIVV`	$\{2^N - 1, 1\}$	fragmented ($N + 1$ sectors)
102	`IVIV`	$\{2^N - 1, 1\}$	fragmented ($N + 1$ sectors)
147	`VVVI`	$\{2^N - 1, 1\}$	$\{2^N\}$
153	`VIVI`	$\{2^N - 1, 1\}$	$\{2^N\}$
195	`VVII`	$\{2^N - 1, 1\}$	$\{2^N\}$

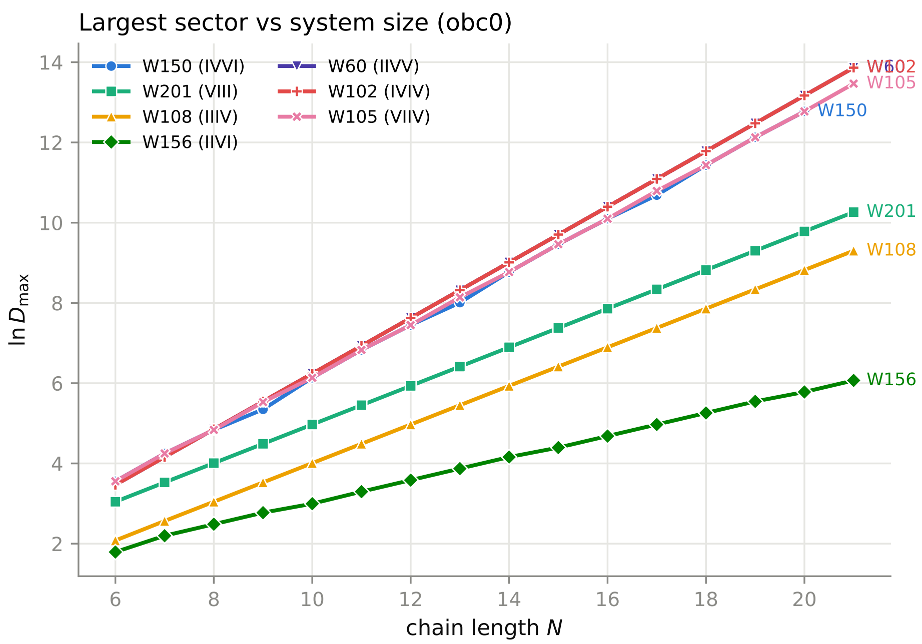

Figure 2: Largest sector $D_{\max}$ vs N (`obc0`), log scale. Slopes give the exponential base: φ (golden ratio) for 201, $4^{1/5} \approx 1.3195$ for 156/198, 2 for the domain-wall family 60/102.

The pattern is not empirical — it follows from the encoding, which is why more data would add nothing. The all-zeros state has every site sitting in neighbour configuration $(0,0)$, so it is frozen precisely when $r_{00} = \texttt{I}$; the all-ones state has every site in configuration $(1,1)$ and is frozen precisely when $r_{11} = \texttt{I}$. Those are the only two configurations that are homogeneous across the whole chain, so no other singleton sector can exist. The table follows immediately: 54/60/102 carry $r_{00} = \texttt{I}$ and freeze the vacuum under both conventions; 147/153/195 carry $r_{11} = \texttt{I}$ and freeze $|1\cdots1\rangle$ under `pbc` only, because the `obc0` padding gives the two end sites a 0 neighbour and breaks the homogeneity; 51/57/99 have neither and are a single sector. The singleton was verified directly at $N = 8$ in each case.

The practical upshot for the scaling tables is that these units are not missing data but a trivial growth law: #sectors is constant (1 or 2) and $D_{\max} = 2^N$ or $2^N - 1$, i.e. exponential with base exactly 2.

5 Boundary-sensitive fragmentation

Two rules change fragmentation class with the boundary:

- **Rules 60/102** (`IIVV`)/(`IVIV`)**:** under `obc0` the domain-wall charge fragments the space into $N+1$ sectors with an extensive dominant sector $D_{\max} = 2^{N-1}$; under `pbc` the periodic seam destroys the charge and the dynamics is ergodic — a single sector of size $2^N - 1$ plus the frozen vacuum $|0\cdots0\rangle$ ($D_{\max}/2^N \to 1$).
- **Rules 150/105:** binomial sector sizes $\binom{N+1}{w}$ under `obc0` (analysed in §6); they *differ* under `pbc`, where the periodic domain-wall counting is rule-specific.

6 Analytic scaling of the binomial rules 150/105

Rules 150 = (`IVVI`) and 105 = (`VIIV`) are the two additive (XOR) elementary rules, $x_i \mapsto x_{i-1} \oplus x_i \oplus x_{i+1}$ and its complement. Under `obc0` their sectors are not merely "central-binomial-like":

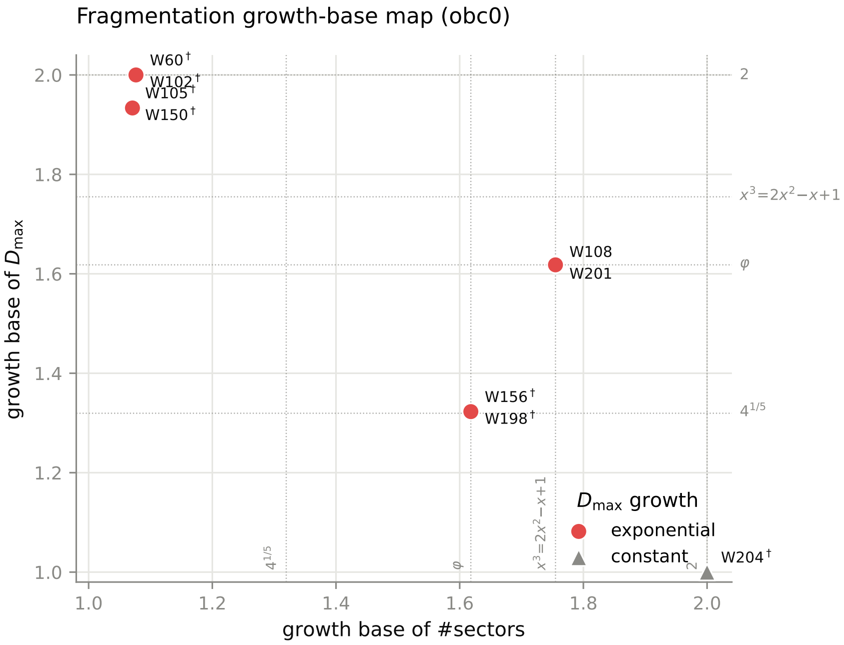

Figure 3: Growth-*base* map (`obc0`): the base $b = e^{\kappa}$ of the sector count against that of the largest sector, one point per rule. Plotting $b \in [1, 2]$ rather than $\kappa \in [0, \ln 2]$ puts the named constants at recognisable positions; the dotted lines mark them. † marks a base from the two-parameter fit $\ln y = c + \kappa N$ rather than an exact recurrence — note in particular that the M2 model $c + \alpha \ln N + \kappa N$ used for the growth *class* is biased low as a rate estimator, reporting 1.281 for 156/198 against its true asymptote $4^{1/5}$.

every sector size is *exactly* a binomial coefficient. Numerically, for all computed $N = 6 \ldots 19$ the recurrent-class sizes are

$$\left\{\binom{N+1}{w}\right\}, \qquad \sum_{\text{sectors}} \binom{N+1}{w} = 2^N, \tag{2}$$

i.e. the 2^N states split into fixed-charge subspaces whose dimensions are the *even-weight* entries of Pascal's row $N + 1$, whose sum is exactly 2^N. The largest sector is therefore

$$D_{\max} = \max_{w \text{ even}} \binom{N+1}{w}, \tag{3}$$

which is the central binomial $\binom{N+1}{\lfloor (N+1)/2 \rfloor}$ except at $N \equiv 1 \pmod 4$, where the central weight is odd and $D_{\max}$ is the near-central entry instead (e.g. $N = 17$: $\binom{18}{8} = 43758$, not $\binom{18}{9} = 48620$) — the same $N \equiv 1$ case treated in the next paragraph.

Now proved, and a sharpened statement (R8). Report R8 derives (2) rather than observing it: in the bond variables $b_j = x_j \oplus x_{j+1}$ the `obc0` map $x \mapsto b$ is a bijection onto the even-weight strings of length $N + 1$ and the elementary move is a hard-core *hop* of one domain wall, so the conserved charge is the open Ising domain-wall number and each wall shell is exactly one sector. That fixes the two places where the phrasing above was loose. First, "even-weight entries" rather than "lower half": the two coincide as multisets only for even N; at odd N the lower half of row $N + 1$ sums to more than 2^N (e.g. $N = 9$: 638 against 512), while the even-weight entries $\{1, 45, 210, 210, 45, 1\}$ are the true sector sizes. Second, the parity caveat in (3). R8 also supplies the exact growth law, $D_{\max}/2^N = 2\sqrt{2/\pi(N + 1)}\,(1 + O(N^{-1}))$, confirming $\alpha = -\frac{1}{2}$ below and settling the fitted base at exactly 2 for all N, not merely over the computed window.

When the two rules agree. For *even* N (so $N+1$ odd) the row has no self-conjugate entry and 150 and 105 give bit-for-bit identical sector multisets. For *odd* N the central weight $(N+1)/2$ is self-conjugate, and the two rules resolve that one sector differently: at $N \equiv 1 \pmod 4$, where $(N+1)/2$ is itself odd, 105 keeps the central binomial whole ($D_{\max} = \binom{N+1}{(N+1)/2}$) while 150 splits it into the conjugate pair $\binom{N+1}{(N\mp1)/2}$ (e.g. $N = 9$: 105 has 252, 150 has $210 + 210$). This does not change the exponents below, since $\binom{N+1}{(N-1)/2} / \binom{N+1}{(N+1)/2} = \frac{(N+1)/2}{(N+3)/2} \to 1$.

Expected exponents. Applying Stirling to (3), $\binom{2m}{m} \sim 4^m/\sqrt{\pi m}$ with $m = \lfloor (N+1)/2 \rfloor$ gives

$$D_{\max} \sim 2^N \sqrt{\frac{8}{\pi(N+1)}}, \qquad \ln D_{\max} = N \ln 2 - \tfrac{1}{2}\ln(N+1) + \tfrac{1}{2}\ln\tfrac{8}{\pi}. \tag{4}$$

So the largest sector is *extensive with base exactly* 2, dressed by a $1/\sqrt{N}$ prefactor. In the report's fit model $\ln y = c + \alpha \ln N + \kappa N$ the prediction is sharp:

$$\kappa = \ln 2 = 0.6931 \text{ (base 2)}, \qquad \alpha = -\tfrac{1}{2}, \qquad c = \tfrac{1}{2}\ln\tfrac{8}{\pi} = 0.467. \tag{5}$$

The number of sectors is the number of admissible charges, $\sim N/2$ — exactly $\lceil (N+2)/2 \rceil$ for 150, and the same up to a small mod 4 wobble for 105. This is a *linear* law ($\kappa = 0$, base 1, power $\alpha \simeq 1$): subexponentially many sectors each of extensive dimension — the signature of a single $U(1)$-like domain-wall charge rather than a shattered spectrum.

Which fit to trust, and the mirror of rule 156. The $-\frac{1}{2}\ln N$ term in (4) is not noise: it is the genuine asymptotic form, so the three-parameter M2 model is the correct estimator here and a bare two-parameter fit $\ln y = c + \kappa N$ is *biased low*. On $N = 6 \dots 19$ the bare fit returns base 1.930 — and it creeps upward with the window (1.904, 1.917, 1.923, 1.930 for $N \le 10, 13, 16, 19$), converging to 2 only as $N \to \infty$, exactly as the $-\frac{1}{2}\ln N/N$ drag predicts. This is the precise *opposite* of rule 156 (§3), where the true law is a pure geometric b^N and it is the M2 $\alpha \ln N$ term that spuriously eats the growth. The moral is that no single fit is universally right: a domain-wall (binomial) mechanism carries a real half-integer α and must be read from M2, whereas a room-packing (transfer-matrix) mechanism is pure-geometric and must be read from the two-parameter fit. Knowing the mechanism, not the residual, selects the estimator.

Two honest finite-size caveats. The fitted α comes out in the range -0.3 to -0.4 (M2 gives -0.31; a fit of $\ln D_{\max} - N\ln 2$ against $\ln N$ gives -0.41) rather than $-\frac{1}{2}$ over this window, because the binomial argument is $N+1$ (not N) and the next Stirling correction $\binom{2m}{m} \propto (1 - \frac{1}{8m})$ adds an $O(1/N)$ term that a three-parameter fit partly absorbs into α; the residual against the *exact* form (4) confirms $\alpha = -\frac{1}{2}$. And κ from M2 reads 0.684 against the exact $\ln 2 = 0.693$ for the same reason — close, and provably converging, but the exact value comes from (3), not from the regression.

7 A unified growth map over all 256 rules

The two preceding sections show why a single "growth base" axis is not enough: 156 and 150/105 both approach the *same* base regime yet obey different laws, and the difference lives entirely in the sub-leading power α. The honest coordinates are therefore the *two* exponents of $y \sim N^\alpha b^N$ — the leading base $b = e^\kappa$ and the sub-leading power α — and plotting both places every growth class on one map without degeneracy (Fig. 4). Exponential rules sit at $b > 1$ with α a small correction; polynomial rules sit at $b = 1$ and climb the α-axis by their degree; a constant law is the origin; an ergodic/volume-law rule is $(b, \alpha) = (2, 0)$. The construction even resolves the two *different* ways of reaching $b = 2$: the binomial rules 150/105 carry $\alpha = -\frac{1}{2}$ and so sit *below* the ergodic point, not on it, while the domain-wall rules 60/102 (with $D_{\max} = 2^{N-1}$) land on it and are separated only in the companion #sectors panel, where their charge shows up as a linear ($\alpha \simeq 1$) sector count.

The points are coloured by rule *family* — *unitary* (`I`/`V` only, 16), *classical* (`V`-free with a reset, 80), and *mixed* (`V` and a reset, 160) — because that is what makes the algebraic bases (un)surprising. A `V`-free rule is deterministic on basis states, so it is an ordinary classical elementary cellular automaton and its sector count is a transfer-matrix count (Fibonacci/Lucas, Padovan/Perrin, tribonacci, …; see R5, "classical baseline"); the striking algebraic bases in the #sectors panel are populated overwhelmingly by these classical rules and by the unitary rules, both classical combinatorics. The *mixed* rules — the genuinely non-classical ones — carry no such clean bases and cluster near $b = 1$ in #sectors while spreading out in $D_{\max}$.

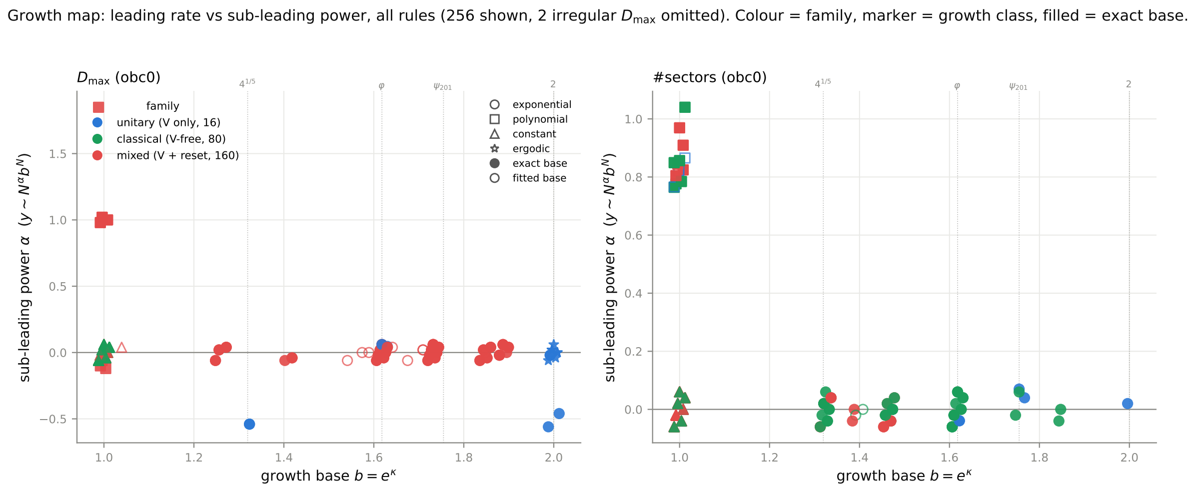

Figure 4: Growth map of all 256 rules in the two exponents of $y \sim N^\alpha b^N$: leading base $b = e^\kappa$ (horizontal) against sub-leading power α (vertical), for $D_{\max}$ (left) and the sector count (right), `obc0`. **Colour** encodes the rule family (blue unitary, green classical/`V`-free, red mixed); **marker** the growth class (○ exponential, □ polynomial, △ constant, ⋆ ergodic); **fill** whether the base is analytically exact (an exact integer recurrence, or a derived unitary constant such as $4^{1/5}$ or the binomial $\alpha = -\frac{1}{2}$) versus a BIC fit. The two *irregular* `obc0` rules (90, 165, whose $D_{\max}$ is a multiplicative order rather than a growth law — see R5) carry no base and are omitted. Reading the colour: the algebraic exponential bases in the #sectors panel are green and blue (classical and unitary combinatorics), while the red mixed rules cluster at $b \approx 1$.

This directly addresses the asymmetry in how the classes are assigned. For the unitary rules the law is analytic: the Fibonacci/tribonacci bases are exact integer recurrences, and 156's $4^{1/5}$ and 150/105's $\alpha = -\frac{1}{2}$ are derived in §§3–6. For the dissipative rules the class comes from BIC with the false-split rate pinned against a null (R5); the map does not pretend these are equally certain, but marks them open. A companion view (Fig. 5) keeps the base of the sector count against the base of $D_{\max}$ on $[1, 2]^2$, with marker size encoding polynomial degree and ergodic rules as stars: it is the more direct way to read *how* a rule fragments — many small sectors (bottom-right, e.g. 204's 2^N singletons), one extensive sector (top-left, the ergodic/domain-wall corner), or both growing together (the diagonal, where 201/108 and 156 live).

Figure 6 folds both views into one joint layout. The main panel is the base-vs-base map of Figure 5; the two margins restore the sub-leading powers of Figure 4 by sharing the main panel's axes. The bottom margin shares the horizontal axis (growth base of #sectors) and plots the #sectors power $\alpha_{\#\text{sec}}$ beneath it; the left margin shares the vertical axis (growth base of $D_{\max}$) and plots the $D_{\max}$ power $\alpha_{D_{\max}}$ beside it. A rule therefore keeps its horizontal position

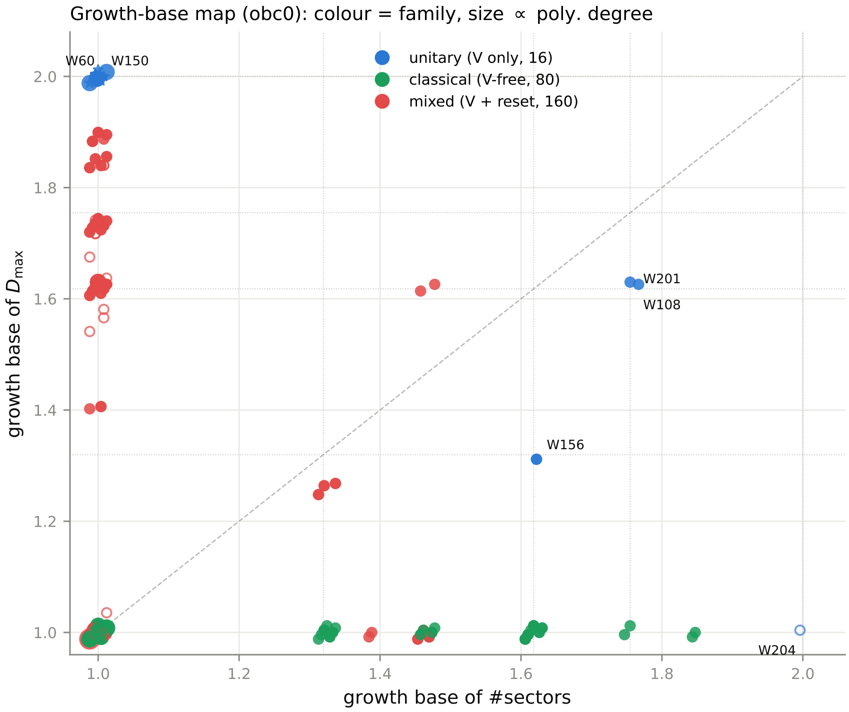

Figure 5: Companion base-vs-base map (`obc0`): growth base of the sector count against that of $D_{\max}$, coloured by family (blue unitary, green classical, red mixed), marker size $\propto$ polynomial degree, ergodic rules as stars. The corners read directly: bottom-right = shattered into $O(2^N)$ frozen sectors (204), top-left = a single volume-law sector (ergodic and the domain-wall family), diagonal = both count and size grow exponentially.

between the main and bottom panels and its vertical position between the main and left panels, so the full (b, α) description of both its series is read from a single point. The classical/unitary rules again occupy the algebraic bases while the mixed rules cluster at $b \approx 1$; the margins add that the sub-exponential structure is a genuine linear ($\alpha \simeq 1$) sector count for the domain-wall rules (bottom margin) and the $-\frac{1}{2}$ binomial prefactor for 150/105 (left margin), neither visible in the base-only view.

8 Broken inversion symmetry (201 vs. 108)

Rules 201 = (`VIII`) and 108 = (`IIIV`) are $0 \leftrightarrow 1$ spin-flip partners. Because the flip conjugates the Hadamard by X and $XHX \neq \pm H$, the symmetry is broken for the full dynamics. At the level of the sector partition it survives under `pbc` (bit-flip is a graph symmetry: $|XHX| = |H|$) but is broken under `obc0` by the vacuum-0 padding: there 201 has Fibonacci $D_{\max}$ and 108 shifted-Fibonacci $D_{\max}$ (matching `HSF` rules 1 and 8). See Report R1 for the exact statement.

9 Reproducibility

All numbers derive from `results/*.jsonl` (engine_version `1a.1`); the table and figures are regenerated by `python -m qca_fragmentation.scaling.summary` and `...scaling.figure`. The sweep is idempotent and checkpointed (`checkpoints/manifest.jsonl`).

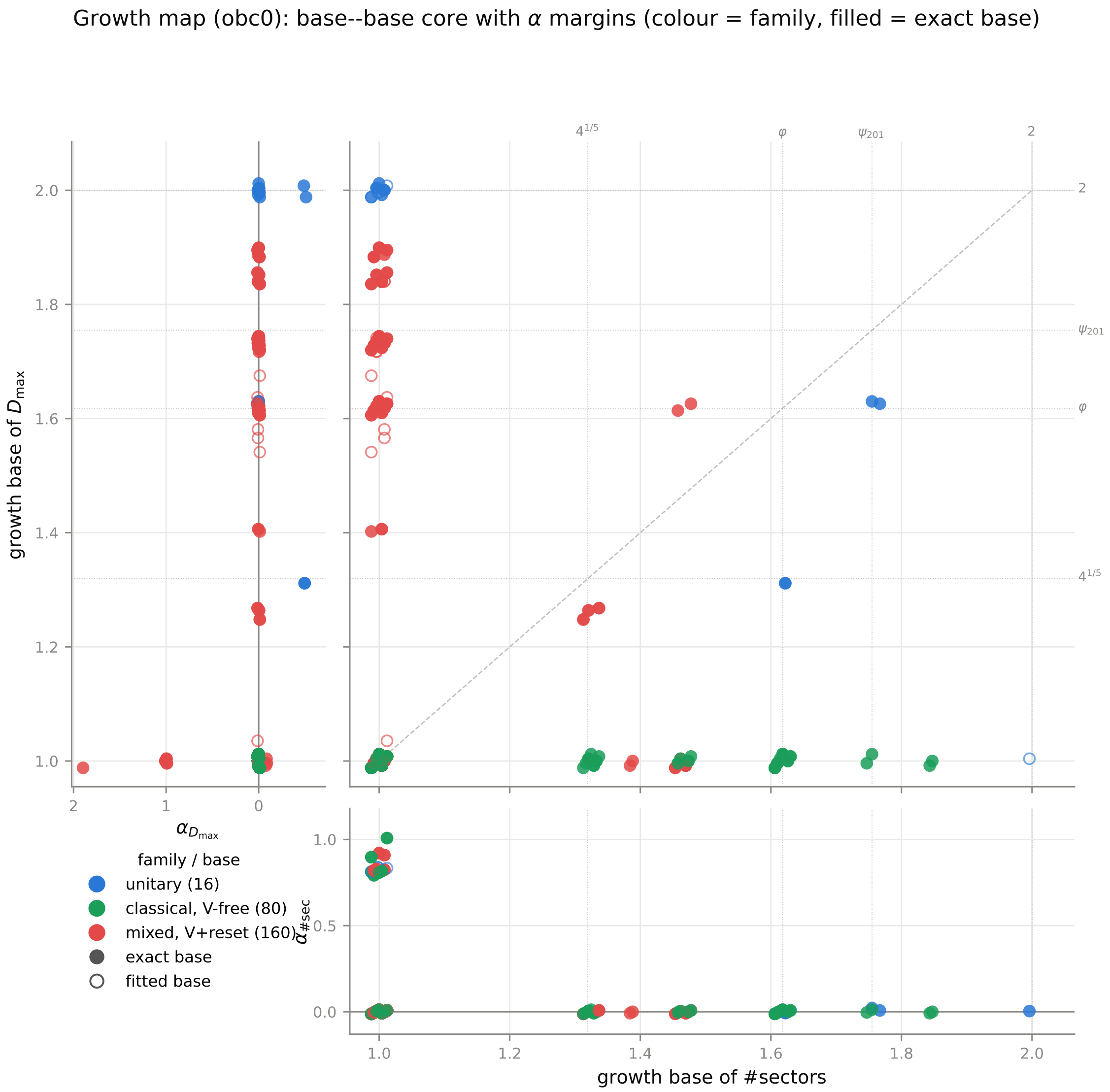

Figure 6: Joint growth map (`obc0`). **Main** panel: growth base of #sectors against growth base of $D_{\max}$ (as Fig. 5). **Bottom** margin (shared x): base of #sectors against its sub-leading power $\alpha_{\#\mathrm{sec}}$. **Left** margin (shared y): the $D_{\max}$ sub-leading power $\alpha_{D_{\max}}$ against the base of $D_{\max}$. Colour is the family, a filled marker an exact base; dotted lines mark the named constants. Points keep their x across the main/bottom panels and their y across the main/left panels.